

# Set 39: Rates of reaction

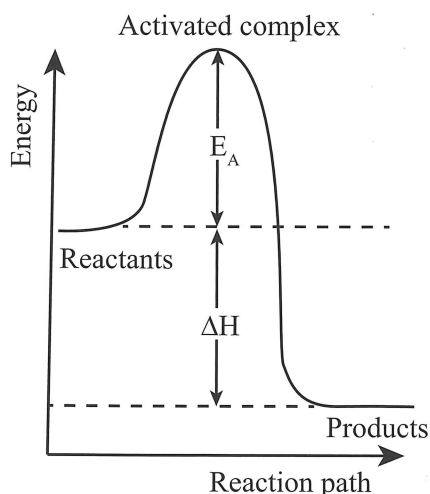
The rates of chemical reactions can be altered in a number of ways, for example, by changing temperature, pressure (for systems involving gases), concentration, degree of sub-division or adding a catalyst.

The collision theory can be used to explain the observed impact of these changes on the rate of a reaction. The collision theory states that, in order for a successful reaction to occur, particles must collide with

- sufficient energy (the activation energy) and
- the appropriate orientation.

Figures 39.1 and 39.2 show energy profile diagrams for an exothermic and an endothermic reaction while figure 39.3 shows the distribution of molecular energies in a system.

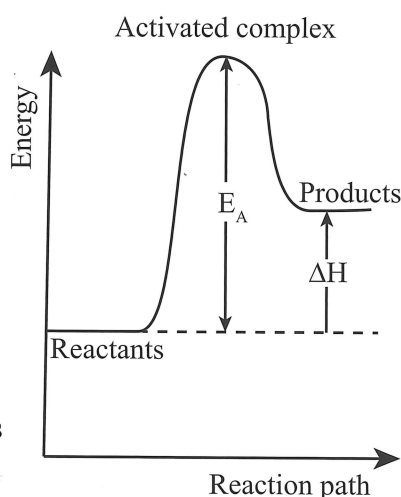
Figure 39.1  
Exothermic reaction



**Exothermic reaction:**

- Energy released to surroundings
- Surroundings warmer
- $\Delta H$  is negative ( $-\Delta H$ )

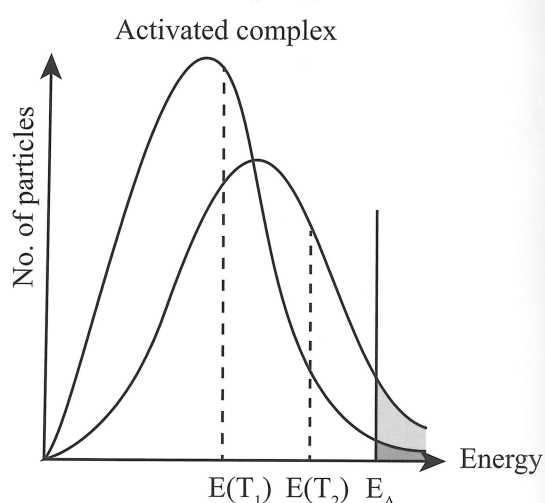
Figure 39.2  
Endothermic reaction



**Endothermic reaction:**

- Energy absorbed from surroundings
- Surroundings cooler
- $\Delta H$  is positive ( $+\Delta H$ )

Figure 39.3  
Molecular energy distribution



$E(T)$  = average energy at temperature  $T$   
 $T_1 < T_2$

The proportion of molecules with energy equal to or greater than the activation energy,  $E_A$  (shaded areas) increases with an increase in temperature

**Notes**

**Example**

Explain why powdered calcium carbonate will react more quickly than the same mass of chips of calcium carbonate when added to dilute hydrochloric acid.

A powdered mass of calcium carbonate will have a much greater surface area than chips. A greater surface area provides more particles available for reaction. As there are more particles, there will be a greater chance of successful collision with hydrogen ions. As there are more successful collisions, there will be a greater rate of reaction.

### Set 39: Exercises

### Notes

1. The following energy values were measured during a reaction.

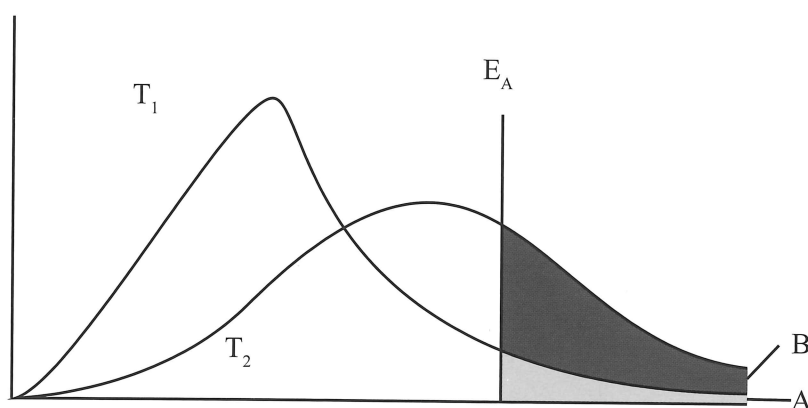
$E$  (reactants) = 110 kJ

$E$  (products) = 130 kJ

$E_A$  = 30 kJ

$\Delta H$  = +20 kJ

- Draw and label an energy profile diagram for the reaction.
  - Is the reaction described endothermic or exothermic? Explain
  - This reaction is a reversible reaction. For the reverse reaction, state the
    - $\Delta H$
    - $E_A$
  - On your diagram for (a), draw in a different colour and label, the reaction pathway for the reaction, if a catalyst was used.
2. The energy distribution diagram shows the distribution of kinetic energy for reacting particles at two different temperatures  $T_1$  and  $T_2$ . Use this diagram to answer the following questions.



- Label the axes.
  - Which is the higher temperature,  $T_1$  or  $T_2$ ? Explain
  - Explain what the line,  $E_A$  represents.
  - What does area A represent?
  - What does area B represent?
3. Industrial processes, in order to be economic, require that products are formed as rapidly as possible at a reasonable cost. The production of ammonia through the Haber process involves an endothermic reaction between nitrogen gas and hydrogen gas to produce ammonia gas.
- Write a balanced equation for the reaction, including the heat of reaction.
  - Explain how increasing the temperature will increase the rate of this reaction.
  - Explain how increasing the pressure of the system will increase the rate of reaction.
  - Explain how the use of a catalyst will increase the rate of reaction.
  - Draw an energy profile diagram for the reaction.

## Set 39: Rates of reaction

### Notes

4. A student was experimenting with ways of increasing the reaction between marble chips (calcium carbonate) and hydrochloric acid.
  - (a) Write a balanced ionic equation for the reaction between calcium carbonate and hydrochloric acid.
  - (b) Give three ways the student could increase the rate of reaction and explain how each change causes the increase.
  - (c) What impact would using acetic acid rather than hydrochloric acid have on the rate of the reaction.
5. When carbon monoxide reacts with oxygen to produce carbon dioxide, 566 kJ of energy is produced per mole of oxygen gas consumed.
  - (a) Write a balanced equation for the reaction, including the heat of reaction.
  - (b) Draw a reaction profile diagram for this reaction given that the activation energy for the reaction is 250 kJ.
  - (c) Annotate the reaction profile diagram to show the impact of adding a catalyst to this reaction.
  - (d) List two other ways of increasing the rate of this reaction and explain how they cause the increase.
6. Flammable liquids do not burn. It is the mixture of the substance's vapour and oxygen that actually ignites. Suggest why it is the vapour that burns and not the liquid itself.
7. List five examples where rates of reaction have been altered in processes occurring in and around the home. Apply the collision theory to describe and explain each of the examples you list.
8. Enzymes are catalysts found in biological systems. Describe how enzymes act as catalysts.
9. When unleaded petrol was introduced catalytic converters were added to cars. These converters are designed to control the emissions produced in the combustion of petrol.
  - (a) Find out which chemicals are used in catalytic converters? Explain how these converters reduce the amount of noxious gases produced by cars. Include equations in your answer.
  - (b) Leaded petrol cannot be used in cars with catalytic converters. Explain why.